Example 1

(a) A modulating signal $m(t) = 10\cos(2\pi \times 10^3 t)$ is amplitude modulated with a carrier signal $c(t) = 50\cos(2\pi \times 10^5 t)$. Find modulation index, the carrier power and the power required for transmitting AM wave.

$m(t) = V_m \cos 2\pi f_m t \rightarrow V_m \cos \omega_m t$

$= 10 \cos(2\pi \times 10^3)t$

$V_m = 10V, \quad f_m = 10^3 Hz$

$c(t) = V_c \cos 2\pi f_c t \rightarrow V_c \cos \omega_c t$

$V_c = 50V, \quad f_c = 10^5 Hz$

$m = \frac{V_m}{V_c}$

$m = \frac{10}{50}$

$\boxed{m = 0.2}$

Example 1

(a) A modulating signal $m(t) = 10\cos(2\pi \times 10^3 t)$ is amplitude modulated with a carrier signal $c(t) = 50\cos(2\pi \times 10^5 t)$. Find modulation index, the carrier power and the power required for transmitting AM wave.

$P_c = \frac{V_c^2}{2R}$

Assuming $R = 1\Omega$

$P_c = \frac{(50)^2}{2} = 1250$ Watts

$P_t = P_c\left(1 + \frac{m^2}{2}\right)$

$P_t = 1250\left(1 + \frac{(0.2)^2}{2}\right)$

$\boxed{P_t = 1275 \text{ Watts}}$

Example 2

The equation of AM wave is given by:

$$\boldsymbol{s(t) = 20[1 + 0.8\cos(2\pi \times 10^3 t)]\cos(4\pi \times 10^5 t)}$$

Find carrier power, total sideband power and bandwidth of AM wave.

Comparing given equation with standard form of AM

$$S(t) = V_c[1 + m\cos(2\pi f_m t)]\cos(2\pi f_c t)$$

$V_c = 20V$

$m = 0.8$

$f_m = 10^3 Hz$

$f_c = 2 \times 10^5 Hz$

$P_c = \frac{V_c^2}{2R} = 200$ watts

$P_t = P_c\left(1 + \frac{m^2}{2}\right) = 264$ watts

Example 2

The equation of AM wave is given by:

$$s(t) = 20\left[1 + 0.8\cos\left(2\pi \times 10^3 t\right)\right]\cos\left(4\pi \times 10^5 t\right)$$

Find carrier power, total sideband power and bandwidth of AM wave.

$$P_{USB} = \frac{m^2 V_c^2}{8R} = 32 \text{ watts}$$

$$P_{LSB} = \frac{m^2 V_c^2}{8R} = 32 \text{ watts}$$

$$P_c = 200 \text{ watts}$$

$$P_{USB} + P_{LSB} + P_c = P_t$$

$$\boxed{P_t = 264 \text{ watts}}$$

Example 2

The equation of AM wave is given by:

$$s(t) = 20\left[1 + 0.8\cos\left(2\pi \times 10^3 t\right)\right]\cos(4\pi \times 10^5 t)$$

Find carrier power, total sideband power and bandwidth of AM wave.

Bandwidth $= 2 f_m$

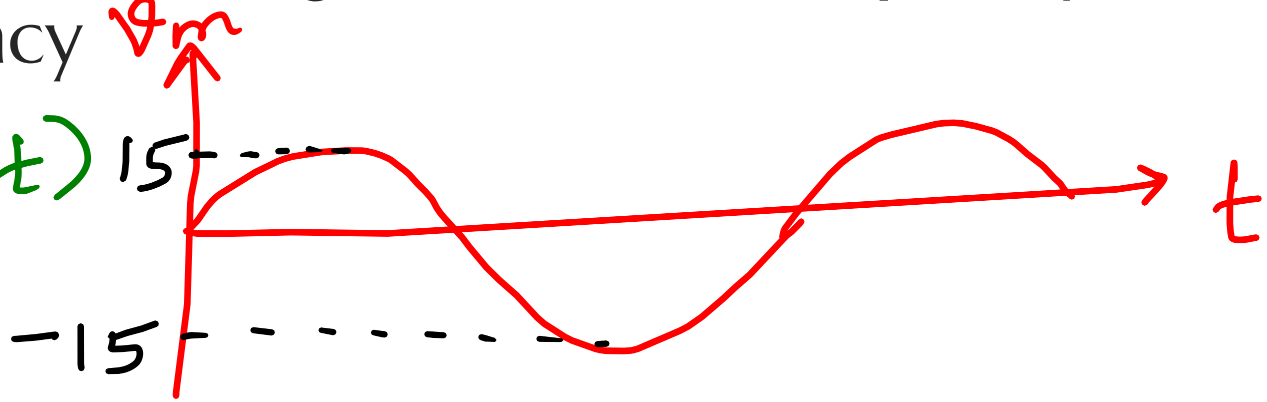
$= 2 \times 10^3$ Hz

$= \boxed{2\text{ kHz}}$

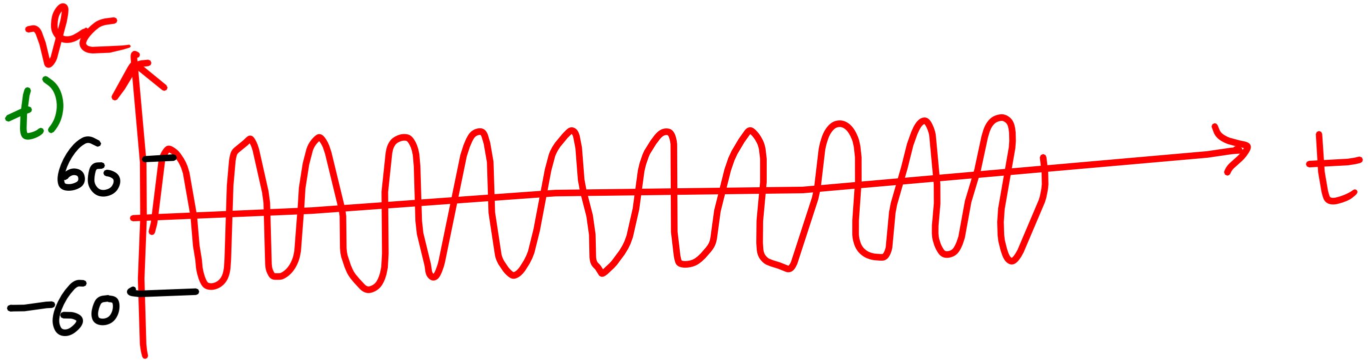
Example 3

An audio signal $15\sin 2\pi(1500t)$ amplitude modulates a carrier given as $60\sin 2\pi(100{,}000t)$. **(i)** sketch audio signal **(ii)** sketch carrier signal **(iii)** sketch waveform of modulated wave **(iv)** fine the modulation index and % of modulation **(v)** what are modulating and carrier frequency **(vi)**Find upper and lower cutoff frequency

message $= 15\sin 2\pi(1500t)$



Carrier $= 60\sin 2\pi(100{,}000t)$



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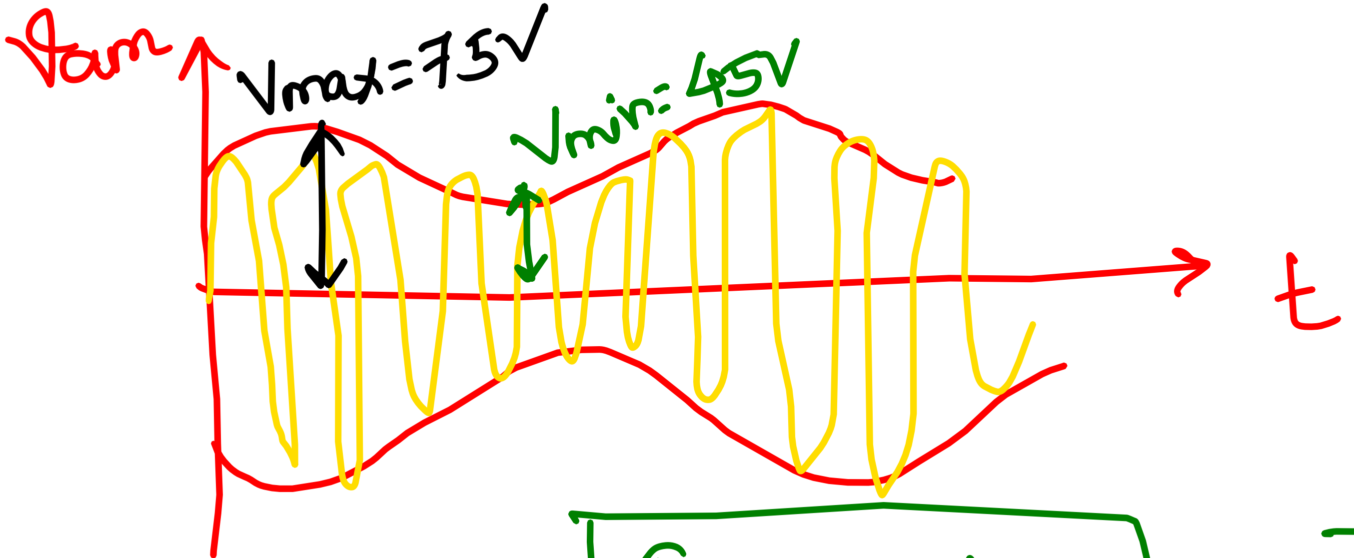


$$m = \frac{V_m}{V_c} = \frac{15}{60} = \boxed{0.25}$$

$$15\sin 2\pi(1500t)$$

$$V_m \sin 2\pi f_m t$$

$$\boxed{f_c = 100\ \text{KHz}}$$

$$\boxed{f_m = 1500\ \text{Hz}}$$

Example 3

An audio signal $15\sin2\pi(1500t)$ amplitude modulates a carrier given as $60\sin2\pi(100{,}000t)$. **(i)** sketch audio signal **(ii)** sketch carrier signal **(iii)** sketch waveform of modulated wave **(iv)** fine the modulation index and % of modulation **(v)** what are modulating and carrier frequency **(vi)**Find upper and lower cutoff frequency

$$f_{USB} = f_c + f_m = 100\times10^3 + 1500\,Hz = 101.5\,KHz$$

$$f_{LSB} = f_c - f_m = 100\times10^3 - 1500\,Hz = 98.5\,KHz$$

Example 4

$x(t) = 10sin\underbrace{4\pi \times 250}_{\omega_m}t$ and $c(t) = 25sin\underbrace{4\pi \times 10^6}_{\omega_c}t$. Calculate:

(i) Modulation index **(ii)** USB and LSB **(iii)** Amplitude of sideband **(iv)** bandwidth **(v)** total power delivered to load of 600Ω.

$V_m = 10V$

$\omega_m = 4\pi \times 250$

$2\pi f_m = 4\pi \times 250$

$f_m = \frac{\cancel{4}^{2}\cancel{\pi} \times 250}{2\cancel{\pi}}$

$\boxed{f_m = 500Hz}$

$V_c = 25V$

$\omega_c = 4\pi \times 10^6$

$2\pi f_c = 4\pi \times 10^6$

$f_c = \frac{4\pi \times 10^6}{2\pi}$

$\boxed{f_c = 2\ MHz}$

$m = \frac{V_m}{V_c} = \frac{10}{25}$

$\boxed{m = 0.4}$

$f_{USB} = f_c + f_m = 2.0005\ MHz$

$f_{LSB} = f_c - f_m = 1.9995\ MHz$

Example 4

$x(t) = 10sin4\pi \times 250t$ and $c(t) = 25sin4\pi \times 10^6t$. Calculate:

(i) Modulation index **(ii)** USB and LSB **(iii)** Amplitude of sideband **(iv)** bandwidth **(v)** total power delivered to load of 600Ω.

$V_{USB} = V_{LSB} = \frac{mV_c}{2} = \frac{0.4 \times 25}{2} = \boxed{5V}$

$BW = 2f_m = 2 \times 500 = 1000Hz$ or $\boxed{1KHz}$

$P_c = \frac{V_c^2}{2R} = 0.52$ watts

$\boxed{P_t = 0.56 \text{ watts}}$

Example 5

A standard AM broadcast station is allowed to transmit modulating frequency up to 5KHz. If the AM station is transmitting on a frequency of 980 KHz, compute the maximum and minimum upper and lower sidebands and total bandwidth.

Example 6

An AM transmitter has a carrier power of 30W. The percentage of modulation is 85%. Calculate: (i) the total power (ii) the power in one sideband

Example 7

An antenna has impedance of 40Ω. An unmodulated AM signal produces a current of 4.8A. The modulation is 90% calculate: (a) the carrier power (b) the total power (c)the sideband power

Example 8

A 400W carrier is modulated to a depth of 75%. Calculate the total power in the modulated wave.

Example 9

A broadcast radio transmitter radiates 10kW. When the modulation percentage is 60. How much of this is carrier power?

Example 10

The antenna current of an AM transmitter is 8A, when only the carrier is sent, but it increases to 8.93A, when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage modulation changes to 0.8

Example 11

A carrier frequency of 5 MHz and peak value of 5V is amplitude modulated by a 4KHz sine wave of amplitude 3V. Determine the modulation index, the upper and lower sideband frequencies, and their amplitude and draw spectrum.

Example 12

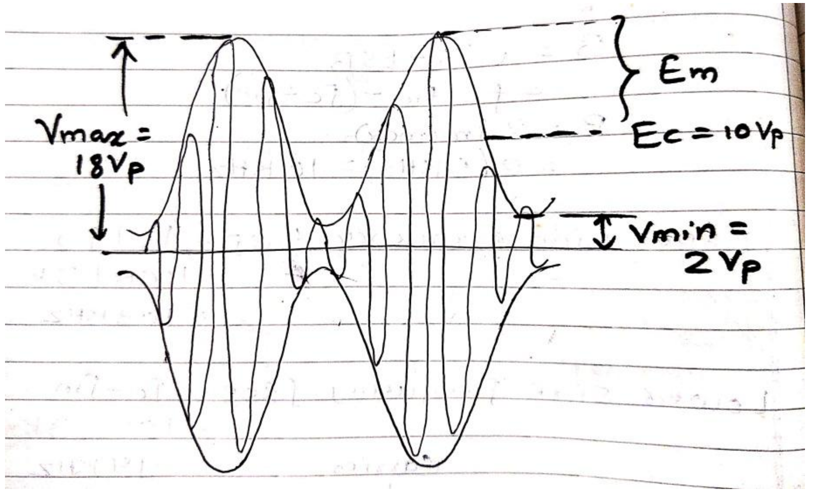
For an AM DSBFC modulator with a carrier frequency of 100KHz and a maximum modulating signal frequency of 5KHz, determine: (i) frequency limits of USB and LSB (ii) Bandwidth (iii) USB and LSB with modulating signal is 3KHz (iv) Draw output frequency spectrum

Example 12

For an AM DSBFC modulator with a carrier frequency of 100KHz and a maximum modulating signal frequency of 5KHz, determine: (i) frequency limits of USB and LSB (ii) Bandwidth (iii) USB and LSB with modulating signal is 3KHz (iv) Draw output frequency spectrum

Example 13

For an AM wave shown, determine (i) peak amplitude of USB and LSB (ii) Peak amplitude of unmodulated carrier (iii) Peak change in the amplitude of the envelope (iv) coefficient of modulation (v)% modulation



Example 14

For an AMDSBFC wave a with peak unmodulated carrier voltage of 10V, a load resistance of 10 ohm, and modulation coefficient of 1 determine:

(i)Powers of the carrier, USB and LSB(ii)Total sideband power (iii) total power of the modulated wave (iv) draw the spectrum (v) repeat steps (i) to (iv) for a modulation index of 0.5